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Counterparty Risk in a CDS Portfolio

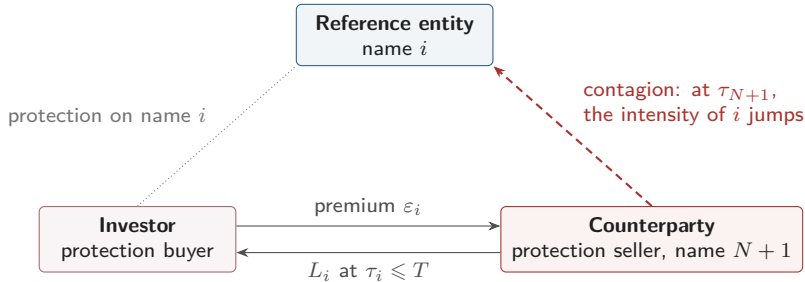
CVA, Netting and Contagion

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A credit default swap, sold by a defaultable counterparty



- **CDS**: the buyer pays the premium rate ε_i until $\tau_i \wedge T$; if the reference entity defaults first, $\tau_i \leqslant T$, the seller pays the loss rate L_i.
- **Contagion**: the default of one name raises the default intensities of the others.
- The portfolio: N such contracts, on names $1, \ldots, N$, with the same seller.

Buying protection from a defaultable seller

- Protection bought on a reference entity from a counterparty: if the counterparty defaults first, it must be bought again, at the spread of that date.
- At the counterparty's default the investor loses the fraction L_{N+1} of the *positive* value of the contract; the **CVA** is the expected discounted loss under $\mathbb{Q}$.
- **Wrong way risk**: the exposure grows as the counterparty's credit quality deteriorates. Here, if its default raises the spread of the reference entity, the protection is worth the most exactly when it is lost, and a CVA computed under independence understates the loss.

"[contagion] might lead to a substantial increase in the credit spread of the reference entity and hence in turn to a much higher replacement value for the CDS"

(McNeil et al., 2015, Section 17.4.4)

The question

In the contagion model of Bo and Ceci (2020), for a portfolio of CDS:

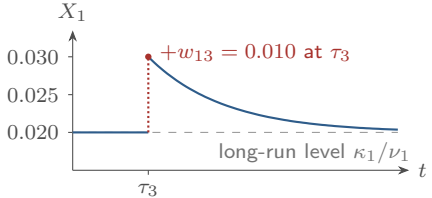
1. how much does contagion change the CVA, against the same model without it?
2. how does it change the benefit of netting the contracts of the portfolio?

The model: interacting default intensities

Names $1, \dots, N$ are the reference entities, $N+1$ the counterparty. Name j defaults at τ_j, with default indicator $H_j(t) = \mathbf{1}_{\{\tau_j \leqslant t\}}$, and $H = (H_1, \dots, H_{N+1})$. The information at t is $\mathcal{G}_t = \sigma\big(W_j(s), H_j(s);\ s \leqslant t,\ j = 1, \dots, N+1\big)$. Under $\mathbb{Q}$, $r = 0$:

$$dX_i(t) = \underbrace{\big(\kappa_i - \nu_i X_i(t)\big)dt + \sigma_i \sqrt{X_i(t)}\, dW_i(t)}_{\text{square-root diffusion}} + \underbrace{\sum_{j \neq i} w_{ij}\, dH_j(t)}_{\text{contagion}}$$

- **Between defaults**: reverting to κ_i/ν_i.
- **At the default** τ_j: the intensity of every surviving name $i \neq j$ jumps by $w_{ij} \geqslant 0$, then decays exponentially at rate ν_i.
- **Independent** $W_1, \dots, W_{N+1}$: the names are linked only through the jumps, so contagion is isolated from correlation. At $w = 0$ the default times are independent.



The path of X_1 at $N = 2$ without its Brownian term, its mean between defaults, from the long-run level; the counterparty defaults at τ_3, and the jump adds half the long-run level.

The price of a CDS: default leg minus premium leg

The CDS on name i, protection bought; with $r = 0$, its price at t is

$$S_i(t,T) = \underbrace{\mathbb{E}^{\mathbb{Q}}\big[L_i\big(H(\tau_i)\big)\,\mathbf{1}_{\{t<\tau_i\leqslant T\}} \;\big|\; \mathcal{G}_t\big]}_{\text{default leg: loss rate } L_i \text{ received at } \tau_i\leqslant T} \;-\; \underbrace{\varepsilon_i\,\mathbb{E}^{\mathbb{Q}}\Big[\int_t^T \mathbf{1}_{\{\tau_i>s\}}\,ds \;\Big|\; \mathcal{G}_t\Big]}_{\text{premium leg: rate } \varepsilon_i \text{ paid until } \tau_i\wedge T}.$$

- **Without contagion**: the premium leg integrates the survival function of τ_i, the default leg its density.
- **With contagion**: the intensity of i jumps at the other names' defaults, and L_i depends on who has defaulted: no survival-function formula.

The pricing function

(X, H) is Markov, with values in $(0,\infty)^{N+1} \times \{0,1\}^{N+1}$: of the history in $\mathcal{G}_t$ only the current state matters. On $\{\tau_i > t\}$

$$S_i(t,T) = F_i^{\text{cds}}\big(t, X(t), H(t)\big),$$

$F_i^{\text{cds}}(t,x,z)$: the same two legs, from $X(t) = x$, $H(t) = z$.

The pricing function: one Cauchy problem per default state

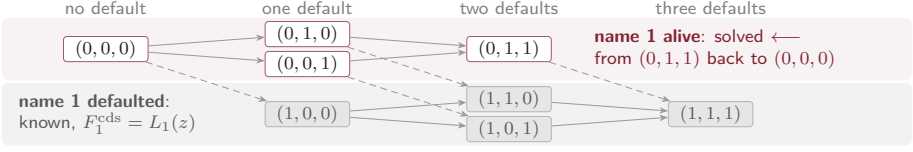
Where name i is alive, $(\partial_t + \mathcal{A})F_i^{\text{cds}} = \varepsilon_i$, $F_i^{\text{cds}}(T,\cdot,\cdot) = 0$, with $\mathcal{A}$ the generator of (X, H):

$$\mathcal{A}f(x,z) = \widetilde{\mathcal{A}}f(x,z) + \sum_{j=1}^{N+1} \big[f(x + w_j, z^j) - f(x,z)\big](1 - z_j)\, x_j,$$

$\widetilde{\mathcal{A}}$ the generator of $\widetilde{X}$, the diffusion without jumps; z_j the j-th component of z, z^j the state z with name j defaulted. The jumps lead only to states with one more default: once those are solved, the problem at z is linear parabolic for $\widetilde{X}$ (**transport form**), and Feynman–Kac applies:

$$\big(\partial_t + \widetilde{\mathcal{A}}\big)u - c_z\, u + f_{i,z} = 0 \quad \Longrightarrow \quad F_i^{\text{cds}}(t,x,z) = \mathbb{E}\Big[\int_t^T f_{i,z}\big(s, \widetilde{X}(s)\big)\, e^{-\int_t^s c_z(\widetilde{X}(u))\, du}\, ds\Big],$$

$\widetilde{X}(t) = x$. Killing rate: the intensities of the names alive, $c_z(x) = \sum_{j:z_j=0} x_j$; source, from the states already solved: $f_{i,z}(t,x) = \sum_{j:z_j=0} F_i^{\text{cds}}(t, x + w_j, z^j)\, x_j - \varepsilon_i$.



The CDS on name 1 at $N = 2$, name 3 the counterparty. Each arrow is a default, and shifts the intensities by a column of w.

The CVA under contagion: the exposure is priced after the jump

The portfolio holds $b_i \in \mathbb{R}$ units of the CDS on name i, bought if $b_i > 0$, sold if $b_i < 0$; (x, z) is the state just before τ_{N+1}.

$$\mathrm{CVA}_{\bar{N}}(t,T) = \mathbb{E}^{\mathbb{Q}}\Big[L_{N+1}\big(H(\tau_{N+1})\big)\, \mathbf{1}_{\{t<\tau_{N+1}\leqslant T\}}\ (\varepsilon_{\bar{N}}(\tau_{N+1},T))^{+} \ \Big|\ \mathcal{G}_t \Big],$$

$$\varepsilon_{\bar{N}}(\tau_{N+1},T) = \sum_{i=1}^{N} b_i \underbrace{(1-z_i)}_{\text{alive: before}} \underbrace{F_i^{\mathrm{cds}}\big(\tau_{N+1},\, x + w_{N+1},\, z^{N+1}\big)}_{\text{price: after the jump}}$$

A Cauchy problem for the CVA

$\mathrm{CVA}_{\bar{N}}(t,T) = g^{\mathrm{cds}}\big(t, X(t), H(t)\big)$, where $(\partial_t + \mathcal{A})g^{\mathrm{cds}} + \Phi = 0$, $g^{\mathrm{cds}}(T,\cdot,\cdot) = 0$ and

$$\Phi = \underbrace{L_{N+1}(z^{N+1})}_{\text{severity}}\ \underbrace{(\varepsilon_{\bar{N}})^{+}}_{\text{at risk}}\ \underbrace{(1-z_{N+1})\,x_{N+1}}_{\text{trigger}}.$$

Derived from the definition of the CVA; the prices are solved first, then g^{cds}.

- **Under independence**, constant loss rate: the default only says *when* the exposure is read,

$$\mathrm{CVA}^{\mathrm{indep}} = L_{N+1}\int_0^T \mathrm{EE}(s)\, f_{N+1}(s)\, ds,$$

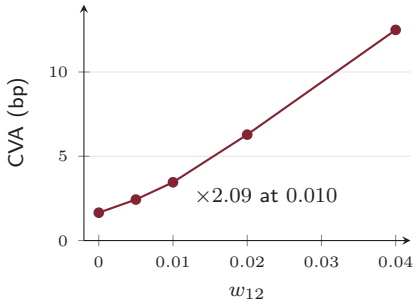
  $\mathrm{EE}(s)$ the expected exposure, f_{N+1} the density of τ_{N+1}.
- **Under contagion** it also changes *what* is read: the exposure is priced after the jump. Wrong way risk, inside the model.

At $N = 1$ the direction of the effect is proved

The direction of the contagion effect

Let $N = 1$, name 2 the counterparty, $L_1((1,1)) \geqslant 0$ and $\varepsilon_1 \geqslant 0$. If $b_1 > 0$ (protection bought), the CVA is non-decreasing in w_{12}; if $b_1 < 0$, non-increasing.

- **Idea**: after the counterparty's default the CDS price is non-decreasing in X_1; w_{12} raises X_1 exactly at that default; the positive part is monotone with the sign of b_1.
- At $w_{12} = 0$: a **lower bound** for the buyer, an upper bound for the seller.
- The direction, not the size; and with one contract netting is invisible: hence $N = 2$.



The validation run at $N = 1$, $w_{21} = 0.008$, $n_x = 61$; the point at 0 is the independence benchmark.

At $N = 2$: the experiment, and a benchmark inside the model

Two reference entities, names 1 and 2, sold by the counterparty, name 3; $T = 5$.

Portfolios: $b = (1, 1)$, protection bought on both; $b = (1, -1)$, bought on one and sold on the other.

Scheme: explicit finite differences, the 2^3 default states solved together, checked against simulation.

Benchmark: $w = 0$ and constant loss rates: independent default times, and the CVA under independence, inside the same model.

$$\bar{w} = \begin{pmatrix} 0 & 0.006 & \mathbf{0.010} \\ 0.005 & 0 & \mathbf{0.009} \\ 0.008 & 0.007 & 0 \end{pmatrix}$$

w_{ij}: the jump of X_i at the default of name j.

Three blocks:

w^{E}, **exposure**: counterparty $\to$ reference entities;

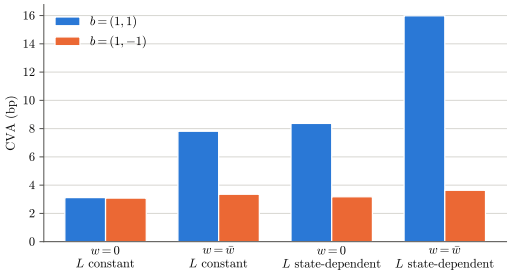
w^{C}, **clustering**: reference entities $\to$ counterparty;

w^{I}, **intra-netting**: between the reference entities.

	L_i constant	L_i state-dependent
$w = 0$	benchmark	severity alone
$w = \bar{w}$	contagion alone	both channels

Loss rates: constant, $L_i = 0.60$; state-dependent, 0.60 plus 0.08 per default of another name.

Contagion raises the CVA of the bought portfolio by 150%



CVA in basis points of the unit notional; $n_x = 33$ points per axis.

$b = (1, 1)$, benchmark 3.12 bp:

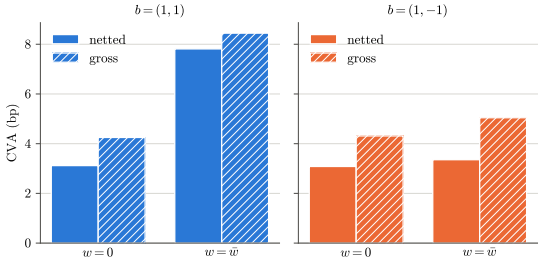
- **contagion**: 7.81 bp, $+\mathbf{150.5}\%$;
- **severity**, independent defaults: 8.37 bp, $+168.3\%$;
- **both**: 15.98 bp, $+412.4\%$, 29% above the sum.

Where: of the $+4.69$ bp of contagion, the exposure block w^{E} alone gives $+4.42$: the only block acting at the counterparty's default.

$b = (1, -1)$: $+8.8\%$, $+3.0\%$, $+18.0\%$; the two price increases largely offset in the netted exposure.

The seller's default raises the spreads of the reference entities when the exposure is priced: the protection is worth more exactly when it is lost.

Contagion erodes netting for one portfolio and strengthens it for the other



Reduction due to netting	$w = 0$	$w = \bar{w}$
$b = (1, 1)$	26.6%	$\mathbf{7.5}\%$
$b = (1, -1)$	28.5%	$\mathbf{33.5}\%$

Constant loss rates. Two portfolios: no general rule is claimed.

Netting

With netting the exposure is the positive part of the portfolio value; without, the sum of the positive parts of the contract values, which gives $\mathrm{CVA}_{\bar{N}}^{\mathrm{gross}}$. A CDS value has either sign, and $(\sum_i x_i)^+ \leqslant \sum_i x_i^+$: so $\mathrm{CVA}_{\bar{N}} \leqslant \mathrm{CVA}_{\bar{N}}^{\mathrm{gross}}$, with equality when the values $b_i S_i$ have a common sign.

- At $w = 0$ the two values are independent and each takes both signs.
- Contagion raises both prices at the counterparty's default: towards a **common** sign for $(1, 1)$, towards **opposite** signs for $(1, -1)$.

Conclusions

Established

- **Method**: under contagion the prices and the CVA are the solutions of recursive Cauchy systems over the default states; the CVA problem derived from its definition, and solved numerically.
- **Direction**: proved at $N = 1$; without contagion, a lower bound for the protection buyer.
- **Size** at $N = 2$: $+150.5\%$ from contagion, $+412.4\%$ with state-dependent loss rates; almost all from w^{E}.
- **Netting**: $26.6\% \to 7.5\%$ for $(1, 1)$, $28.5\% \to 33.5\%$ for $(1, -1)$.

Limits

- Illustrative parameters, no market data; the weights varied, not estimated.
- At $N = 2$ the results are observed in the configurations computed, not proved.
- Convergence of the scheme not established; its error measured.
- Computational cost: 2^{N+1} grids in $N + 1$ dimensions, hence $N \leqslant 2$; $r = 0$.

Extensions

Calibration to CDS spreads: do the quoted spreads determine the contagion weights? Hedging the CVA, as in the reference paper.

Thank you

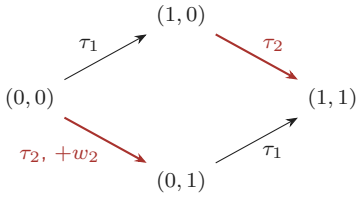
Backup: the pricing system in full

At each default state z, the price of the CDS on name i solves

$$\Big(\frac{\partial}{\partial t}+\mathcal{A}\Big)F_i^{\mathrm{cds}}-(1-z_i)\,\varepsilon_i-z_i\sum_j\big[L_i(z^j)-L_i(z)\big](1-z_j)\,x_j=0,\qquad F_i^{\mathrm{cds}}(T,x,z)=L_i(z)\,z_i.$$

- The last term compensates the jumps of a state-dependent loss rate; it is switched on only after the default of name i, where the price is the constant $L_i(z)$.
- At a fixed state: a linear parabolic problem for the diffusion without jumps, with killing rate $c_z(x)=\sum_{j:z_j=0}x_j$, the total rate of leaving z, and a source built from the successor prices read at $x+w_j$, times the default rates, less the premium.
- The jump part of the generator is the only place the weights enter, and it moves the intensities and the default state together: dropping the shift $x+w_j$ would keep which names have defaulted and discard the size of the contagion.
- The CVA function g^{cds} solves the same kind of system, with the source Φ; it is linear in its unknown, and the prices enter it only through the source.

Backup: the four states at $N = 1$



Name 1 the reference entity, name 2 the counterparty.

- $(1,1)$ and $(1,0)$: the constants $L_1((1,1))$ and $L_1((1,0))$.
- $(0,1)$: the price of a CDS on name 1 alone, $\mathbb{E}\big[\int_t^T \big(L_1((1,1))\,\widetilde{X}_1(s) - \varepsilon_1\big) e^{-\int_t^s \widetilde{X}_1(u)\,du}\,ds\big]$.
- $(0,0)$: reads the price at $(0,1)$ at the shifted point $x_1 + w_{12}$.

The CVA vanishes at three states, and at $(0,0)$

$$g^{\mathrm{cds}}\big(t,x,(0,0)\big) = L_2\big((0,1)\big)\mathbb{E}\Big[\int_t^T \widetilde{X}_2(s)\,\Big(b_1 F_1^{\mathrm{cds}}\big(s,\widetilde{X}(s)+w_2,(0,1)\big)\Big)^+ e^{-\int_t^s (\widetilde{X}_1+\widetilde{X}_2)(u)\,du}\,ds\Big]:$$

the discount is the joint survival of both names; the exposure is read after the jump, the counterparty's default rate before it. Only w_{12} reaches the CVA.

Backup: the benchmark without contagion

The benchmark

With $w = 0$ and constant loss rates, each price is the price of a CDS on its name alone, $\varphi_i(t, x_i)$, and

$$\mathrm{CVA}_{\bar{N}}(0,T) = L_{N+1} \int_0^T \mathbb{E}^{\mathbb{Q}}\Big[\Big(\sum_{i=1}^{N} b_i \, \mathbf{1}_{\{\tau_i > s\}} \, \varphi_i\big(s, X_i(s)\big)\Big)^+\Big] \, f_{N+1}(s)\, ds,$$

the CVA under independence.

- The comparison is *not* the inequality $\mathrm{CVA} > \mathrm{CVA}^{\mathrm{indep}}$ of the wrong way risk literature: setting $w = 0$ changes the laws of the default times and the prices, not only their dependence.
- Constant loss rates at $w \neq 0$ isolate contagion; state-dependent loss rates at $w = 0$ isolate severity.
- Premium rates at par for $w = 0$, $L_i = 0.60$: 119.08 and 148.69 bp, then fixed in every configuration.

Backup: the scheme and its validation

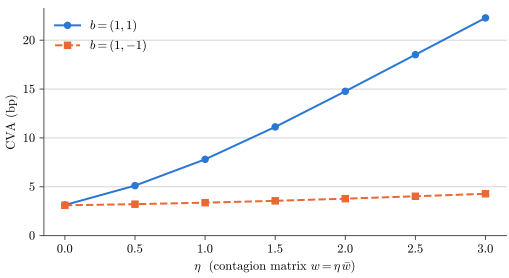
$$u_z^{k+1} = u_z^k + \Delta t\Big(\widetilde{\mathcal{A}}_h u_z^k - c_z\, u_z^k + \sum_{j\,:\,z_j=0} \mathcal{I}_h u_{z^j}^k(x + w_j)\, x_j + \vartheta_z^k\Big)$$

- **Derived**: under the time-step bound the coefficients are non-negative: monotone, bounded in the maximum norm, CVA $\geqslant 0$.
- **Not established**: convergence, and its rate.
- **Measured**: at $n_x = 25, 33, 41$ the changes are in the ratio 0.462 (second order predicts 0.463); error at 33 points 0.0179 bp (0.23%); simulation within 1.6 standard errors.

w	b	FD	MC, 4000 steps
0	$(1,1)$	3.1188	3.0991 ± 0.0124
	$(1,-1)$	3.0833	3.0836 ± 0.0110
$w^{\mathrm{E}} + w^{\mathrm{C}}$	$(1,1)$	7.5695	7.5416 ± 0.0204
	$(1,-1)$	3.3147	3.3158 ± 0.0114

The simulation needs the price at the counterparty's default in closed form, which holds when $w^{\mathrm{I}} = 0$.

Backup: the level of contagion, and the three blocks



With $w = \eta\,\bar{w}$, $\eta = 0, \ldots, 3$: from 3.13 to 22.28 bp for $b = (1, 1)$, from 3.10 to 4.28 bp for $(1, -1)$; non-decreasing at every value computed.

Blocks switched on one at a time, $b = (1, 1)$:

w	CVA (bp)	increment
0	3.1188	—
w^{E}	7.5383	$+\mathbf{4.4195}$
w^{C}	3.1355	$+0.0167$
w^{I}	3.2533	$+0.1345$
$\bar{w}$	7.8133	$+4.6945$

w^{E}: in the source of the CVA, through the shift $x + w_3$. w^{I}: one default later. w^{C}: only after a reference entity has defaulted, on the counterparty's intensity.

Not additive: one at a time the increments sum to 4.5708 bp, against 4.6945 jointly; w^{I} adds 0.1345 bp alone and 0.2438 bp on top of $w^{\mathrm{E}} + w^{\mathrm{C}}$.

Backup: the counterparty's loss rate

The counterparty's loss rate is held constant, since it multiplies the exposure without making it depend on the counterparty's default. Made state-dependent, it adds

- for $b = (1, 1)$: 0.0718 bp ($w = 0$) and 0.1403 bp ($w = \bar{w}$), 1.3% and 1.7% of the increase;
- for $b = (1, -1)$: 0.0400 and 0.0763 bp, 30.1% and 21.3% of it, the increases of the two contracts offsetting each other.

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